

Lab 3

Working with Gradle: Setting Up a Gradle Project, Understanding Build Scripts (Groovy and Kotlin DSL), Dependency Management and Task Automation.

Create folder in desktop with name “gradle-project” inside that create folder with name “groovy”

Step 1: Create a new Project

Cd path of groovy folder

```
gradle init --type java-application
```

Step 2: build.gradle (Groovy DSL)

gradle-project -> groovy -> app -> build.gradle -> default code will be generated, compare and add the required line.

```
plugins {  
    id 'application'  
}  
  
application {  
    mainClass = 'com.yit.AdditionOperation'  
}  
  
repositories {  
    mavenCentral()  
}  
  
dependencies {  
    testImplementation 'junit:junit:4.13.2'  
}  
  
test {  
    outputs.upToDateWhen { false }  
}
```

```

testLogging {
    events "passed", "failed", "skipped"
    exceptionFormat "full"
    showStandardStreams = true
}
}

```

Step 3: AdditionOperation.java(Change file name and update below code)

- After creating project **change the file name**.
- Inside gradle-project, manually navigate the folder path like **app/src/main/java/com/yit/**
- Change the file name **App.java** to **AdditionOperation.java**
- After then open that file and type the code for Addition of two numbers.

```

package com.example;

public class AdditionOperation {
    public static void main(String[] args) {
        double num1 = 5;
        double num2 = 10;
        double sum = num1 + num2;
        System.out.printf("The sum of %.2f and %.2f is %.2f\n", num1, num2, sum);
    }
}

```

Step 4: AdditionOperationTest.java (JUnit Test) (Change file name and update below code)

- After creating project **change the file name**.
- Inside gradle-project, manually navigate the folder path like **app/src/test/java/com/example/**
- Change the file name **AppTest.java** to **AdditionOperationTest.java**
- After then open that file and type the code for Addition of two numbers.

```

package com.example;

import org.junit.Test;

import static org.junit.Assert.*;

```

```

public class AdditionOperationTest {

    @Test

    public void testAddition() {

        double num1 = 5;

        double num2 = 10;

        double expectedSum = num1 + num2;

        double actualSum = num1 + num2;

        assertEquals(expectedSum, actualSum, 0.01);

    }

}

```

Step 5: Run Gradle Commands

- To **build** the project:

C:\Users\USER\Desktop\gradle-project> groovy>gradle build

- To **run** the project:

C:\Users\USER\Desktop\gradle-project>groovy>gradle run

- To **test** the project:

C:\Users\USER\Desktop\gradle-project>groovy>gradle test

Gradle Kotlin

In the desktop, inside gradle-project folder, create folder with name kotlin

Step 1: Create a new Project

Cd path of kotlin folder

Step 1: Create a new Project

gradle init --type java-application

Copy the src folder and build.gradle.kts outside app folder and delete app folder

Change the C:\Users\USER\Desktop\gradle-project\kotlin\src\main\java\com\yit

With C:\Users\USER\Desktop\gradle-project\kotlin\src\main\kotlin\com\yit

Step 3: Main.kt (Change file name and update below code)

- After creating project **change the file name**.
- Manually navigate the folder path like **src/main/kotlin/com/yit/**
- Change the file name **App.java** to **Main.kt**
- After then open that file and write the code for Addition of two numbers.

```
package com.yit
```

```
fun addNumbers(num1: Double, num2: Double): Double { // ← Top-level  
    return num1 + num2  
}
```

```
fun main() {  
    val num1 = 10.0  
    val num2 = 5.0  
    val result = addNumbers(num1, num2) // Can call from anywhere  
    println("The sum of $num1 and $num2 is: $result")  
}
```

Step 4: MainTest.kt (JUnit Test) (Change file name and update below code)

- After creating project **change the file name**.
- Manually navigate the folder path like **src/test/kotlin/com/yit/**
- Change the file name **MainTest.java** to **MainTest.kt**
- After then open that file and write the code for Addition of two numbers with Test block.

```
package com.yit
```

```
import org.junit.Test  
import org.junit.Assert.assertEquals
```

```
class MainTest {  
  
    @Test  
    fun testAddNumbers() {  
        val num1 = 10.0  
        val num2 = 5.0
```

```
val result = addNumbers(num1, num2)

assertEquals(
    "The sum of $num1 and $num2 should be 15.0",
    15.0,
    result,
    0.001
)
}
}
```

➤ **Step5: make the changes to default generated build.gradle.kts file**

```
plugins {
    kotlin("jvm") version "1.9.24"
    application
}

group = "com.yit"
version = "1.0-SNAPSHOT"

repositories {
    mavenCentral()
}

dependencies {
    implementation(kotlin("stdlib"))
    testImplementation("junit:junit:4.13.2")
}

application {
    mainClass.set("com.yit.MainKt")
}

tasks.test {
```

```
useJUnit()

testLogging {
    events("passed", "failed", "skipped")
    exceptionFormat = org.gradle.api.tasks.testing.logging.TestExceptionFormat.FULL
    showStandardStreams = true
}

outputs.upToDateWhen { false }
}

java {
    toolchain {
        languageVersion.set(JavaLanguageVersion.of(17))
    }
}
```

Step 5: Run Gradle Commands

gradle build

- To **run** the project:

gradle run

- To **test** the project:

gradle test